

Experiment: *Exploring Application Deployment and Release Automation (ARA) in DevOps using Free ARA Tools*

In DevOps, ARA plays a crucial role in enabling Continuous Delivery (CD) by automating the build, deployment, and release of applications with minimal manual intervention.

Task:

Using **free and open-source ARA tools**, design and perform a simple experiment to demonstrate how deployment automation supports **CD**.

Perform the following:

1. Create a simple application (e.g., a basic HTML page or “Hello World” web app).
2. Use **Git** (free) as a version control system to store the application source code.
3. Configure **Jenkins (Community Edition)** or **GitHub Actions** to automatically build the application whenever a code change is pushed to the repository.
4. Use **Docker** to package the application into a container image.
5. Automate the deployment of the containerized application to a local environment using **Docker** or **Docker Compose**.
6. Trigger the entire pipeline automatically on every code change.
7. Observe how automation reduces manual effort and speeds up the release process.

Questions to Answer:

1. How does Application Deployment and Release Automation improve the reliability and speed of software delivery?
2. Explain how the automated pipeline implemented in this experiment supports **Continuous Delivery**.
3. Identify the role of each ARA tool used in the experiment.
4. What challenges might arise if deployment and release were done manually?

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.

2. The report should contain code(s), step-by-step procedure followed, output snapshots and any other asked details.
3. The PDF name should be in the form "StudentName_SAPID".